BiblioBlitz v3.2

Global Open-Access Academic Paper Downloader

User & Technical Documentation

Ayanava Poddar

DOI: 10.5281/zenodo.20469653

GitHub: https://github.com/Ayanava-23556003/BiblioBlitz

License: MIT | Language: Python 3.9+ | Platform: Windows / macOS / Linux

May 2026

1 Overview

BiblioBlitz is a free, open-source, cross-platform desktop application that automates the discovery and download of open-access academic papers from any field of research. It aggregates results from five major bibliographic APIs — CrossRef, OpenAlex, Semantic Scholar, PubMed/NCBI, and CORE — deduplicates by DOI, allows the user to select specific journals from a live-fetched list, filters by country of author affiliation, supplements open-access links via Unpaywall, and downloads PDF files automatically.

BiblioBlitz requires no installation of R, PowerShell, or any scientific computing framework on the end-user machine. A self-contained Windows executable is provided; source code runs directly on macOS and Linux with Python 3.9+.

1.1 Version History

Version	**Key Changes**
v2.0 (PaperFinder)	R and PowerShell backend; CrossRef only; Windows 64-bit; hardcoded Q1 list
v3.0 (BiblioBlitz)	Pure Python; 5 APIs; all journals worldwide; smart placeholders; bundled logo; exit confirmation
v3.1	Journal selector dialog; country/region filter (80+ countries); OpenAlex live journal fetch
v3.2	Bug fixes (NoneType crash, OpenAlex country filter); ICO icon support; privacy documentation

2 System Requirements

2.1 End Users (running the EXE)

- Windows 10 or Windows 11 (32-bit or 64-bit)
- Internet connection (required for API searches and PDF downloads)
- No Python, R, or any other runtime required

2.2 Developers (building from source)

- Python 3.9 or later
- pip packages: customtkinter, Pillow, pyinstaller
- Internet connection
- Windows recommended for building the EXE (build_exe.bat)

3 Installation

3.1 End Users — Windows EXE

1. Download BiblioBlitz_v3.2.exe from the GitHub Releases page: https://github.com/Ayanava-23556003/BiblioBlitz/releases
2. Double-click the file to launch. No installation wizard is required.
3. If Windows Defender SmartScreen appears, click 'More info' then 'Run anyway'. This is normal for unsigned executables distributed outside the Microsoft Store.

3.2 Developers — Running from Source

1. Ensure Python 3.9+ is installed and added to PATH.
2. Install dependencies: open a terminal and run:

   ```
   pip install customtkinter Pillow
   ```

3. Run the application:

   ```
   python biblioblitz.py
   ```

3.3 Building the EXE

4. Place biblioblitz.py, build_exe.bat, and the logo PNG in the same folder.
5. Double-click build_exe.bat.

6. The script installs dependencies and runs PyInstaller automatically.
7. The output BiblioBlitz_v3.2.exe will appear in the dist\ subfolder.
8. The EXE can be copied to any Windows PC and run without any prerequisites.

4 User Guide

4.1 Interface Overview

The BiblioBlitz interface is divided into three areas: the header bar at the top showing the app name, version, and status; the configuration panel on the left where all search settings are entered; and the activity log on the right which shows real-time progress and results.

4.2 Step-by-Step Workflow

Step 1 — Enter Email Address. Type your email address in the Email Address field. This is sent to CrossRef and Unpaywall as required by their fair-use policies. It is not stored by BiblioBlitz.

Step 2 — Set Download Directory. Click the '···' button to browse and select a folder where downloaded PDFs will be saved. The default is a 'Papers' folder in your home directory.

Step 3 — Enter Keywords. Type your search keywords in the Keywords field, separated by commas. Example: soil erosion, runoff, sediment yield. These are used both to search the APIs and to filter paper titles.

Step 4 — Fetch Journals. Click 'Fetch Journals for These Keywords'. BiblioBlitz queries CrossRef and OpenAlex with your keywords and builds a live list of all journals that publish on this topic. This may take 30–60 seconds.

Step 5 — Select Journals. Click 'Select Journals' to open the journal selector dialog. Use the search bar to filter journals by name. Check the journals you want to include. If you leave all unchecked, papers from all journals will be downloaded.

Step 6 — Set Region / Country Filter. Choose a country from the dropdown to restrict results to papers whose authors are affiliated with institutions in that country. Choose 'Global (All Countries)' to search without geographic restriction.

Step 7 — Set Max Papers and Starting Year. Use the sliders or type directly in the entry boxes. Maximum papers: 100 to 1,00,000. Starting year: 1990 to 2025.

Step 8 — Start Download. Click 'Start Download'. The activity log will show real-time progress through all pipeline stages. The statistics bar at the top of the log shows counts for fetched, filtered, open-access, downloaded, skipped, and error papers.

Step 9 — Review Results. Downloaded PDFs are saved in the chosen directory. A download_log.csv file is saved in the same directory, listing every paper with its title, DOI, journal, year, source API, country filter used, download status, and filename.

4.3 PDF Integrity Check

After downloading, click 'Run PDF Integrity Check' to scan the download folder. BiblioBlitz reads the first four bytes of every PDF file and verifies the PDF magic number (%PDF). Files that fail this check are moved to a 'Corrupted_PDFs' subfolder. A pdf_integrity_report.csv is saved in the download folder with the status and size of every file checked.

4.4 Stopping a Download

Click the 'Stop' button at any time to interrupt the download gracefully. The current file will finish downloading before the process halts. When the application is closed during a download, a confirmation dialog asks whether to stop and exit.

5 Search Sources and APIs

BiblioBlitz aggregates results from five bibliographic APIs, then supplements open-access links via a sixth API. All are free, official, and legal. Table 2 summarises the sources.

Table 2. Bibliographic APIs used by BiblioBlitz.

API	Coverage	Primary Contribution
CrossRef	150M+ works	Primary journal article metadata; DOI backbone
OpenAlex	250M+ works	Modern open index; built-in OA signals; journal data
Semantic Scholar	200M+ papers	AI-enriched metadata; interdisciplinary coverage
PubMed / NCBI	35M+ records	Biomedical, environmental health, life sciences
CORE	200M+ OA docs	Direct open-access PDF download links
Unpaywall	50M+ OA links	Best-available PDF URL by DOI (supplemental stage)

All API requests are throttled at 0.25–0.5 seconds between calls to comply with fair-use policies. BiblioBlitz identifies itself in every request via a User-Agent header and, where required, a mailto parameter.

6 Processing Pipeline

Table 3. BiblioBlitz 7-stage processing pipeline.

Stage	Name	Description
1	Journal Fetch	CrossRef and OpenAlex queried with keywords; unique journal names collected into a live list for user selection
2	Multi-source Search	All five APIs queried simultaneously with keywords, year filter, and optional country filter
3	Merge & Deduplicate	All results merged; duplicates removed by normalised DOI; PDF URLs transferred to retained records
4	Journal Filter	If journals selected, retains only papers from matching journals; otherwise all papers retained
5	Keyword Filter	Retains only papers where the title contains at least one user keyword

Stage	Name	Description
6	Unpaywall Supplement	For papers without a PDF URL, Unpaywall queried by DOI to find best open-access link
7	PDF Download	PDFs downloaded; CSV log saved; files named as Title [DOI].pdf

7 Journal Selector

The journal selector is a key feature of BiblioBlitz v3.1+. Rather than using a static hardcoded list of journals, BiblioBlitz dynamically fetches journal names from CrossRef and OpenAlex based on your keywords. This means the journal list is always relevant to your research topic and covers every discipline.

The selector dialog provides a search bar for filtering journals by name, Select All and Deselect All buttons, and a counter showing how many journals are currently selected. Selections are remembered until the next journal fetch. If no journals are selected when the download starts, all journals returned by the APIs are included.

8 Country / Region Filter

The country filter allows researchers to restrict results to papers from specific national research communities. This is useful for systematic reviews of national-level research output, policy analyses, or collaboration mapping.

When a country is selected, BiblioBlitz passes the country name or code to the APIs as follows:

- CrossRef: country name appended to the keyword query string
- OpenAlex: authorships.institutions.country_code filter using ISO 2-letter code
- PubMed: country name appended as an affiliation filter
- Semantic Scholar and CORE: country name appended to the keyword query

Note that country filtering is approximate — papers with mixed international authorship may or may not appear depending on the API's affiliation data. 'Global (All Countries)' disables the filter entirely.

9 Output Files

9.1 PDF Files

Each downloaded PDF is saved with the filename format:

```
Paper_Title_Truncated_To_70_Chars
[DOI_with_slashes_replaced].pdf
```

9.2 Download Log (download_log.csv)

A CSV file saved in the download directory after each run, containing the following columns:

Column	Description
Title	Full paper title
Doi	Digital Object Identifier
Year	Publication year
journal	Journal name as returned by the source API
country_filter	Country filter applied during this download run
source	API source that returned this paper (CrossRef, OpenAlex, etc.)
status	success / already_exists / error
File	Filename of the downloaded PDF

9.3 PDF Integrity Report (pdf_integrity_report.csv)

Generated by the PDF Integrity Check, containing FileName, Status (OK / InvalidHeader / TooSmall / CannotOpen), and SizeMB for every PDF in the download folder.

10 Privacy and Data Handling

BiblioBlitz is designed with user privacy as a core principle. The following summarises what data is sent and to whom.

Data	Sent To	Purpose	Stored by BiblioBlitz?
Email address	CrossRef, Unpaywall	Required by fair-use policy for API identification	No
Keywords	All 5 search APIs	Paper discovery	No
Country selection	CrossRef, OpenAlex, PubMed, CORE	Affiliation filter	No
Journal selections	Not sent externally	Local filtering only	No

- No data is ever sent to the BiblioBlitz developers or any third party beyond the APIs listed above.
- No usage analytics, telemetry, or tracking of any kind is collected.
- All processing happens locally on the user's machine.
- Downloaded PDFs are saved only to the folder chosen by the user.

11 Troubleshooting

Problem	Likely Cause	Solution
OpenAlex returns 0 results	Country code filter mismatch	Update to v3.2 which fixes the authorships.institutions.country_code filter

Problem	Likely Cause	Solution
NoneType crash after deduplication	Paper with missing journal name	Update to v3.2 which adds null-check on journal names before filtering
Logo shows as 'O' icon in taskbar	No .ico file provided	Convert PNG to ICO and add iconbitmap() call in __init__; bundle with --add-data in build_exe.bat
Semantic Scholar returns 0 results	API rate limit or temporary outage	Wait and retry; SS has stricter rate limits than other APIs
CORE returns few results	No API key (public rate limit)	Results are limited without a CORE API key; acceptable for moderate use
PDF download fails (FAIL in log)	Paper not freely available at that URL	Check Unpaywall directly at unpaywall.org; some PDFs are behind temporary embargoes
SmartScreen warning on EXE	EXE is unsigned	Click 'More info' then 'Run anyway'; this is normal for unsigned academic software
Build fails — customtkinter not found	pip installed to wrong environment	Run: python -m pip install customtkinter Pillow pyinstaller

12 Known Limitations

- Open-access availability is limited by publisher embargo policies. Papers behind paywalls cannot be downloaded even if they appear in search results.
- Country filtering is approximate and depends on affiliation data quality in each API. International collaborations may not filter cleanly.
- Semantic Scholar and CORE occasionally return incomplete metadata, which may cause papers to be excluded by the keyword title filter.
- The CORE API public rate limit is low; a CORE API key (free from core.ac.uk) will improve CORE result counts in future versions.

- Journal fetching (Stage 1) queries only the first 500 CrossRef records and 100 OpenAlex records. Rare journals may not appear in the selector.
- Large downloads (50,000+ papers) may take several hours and should be left running overnight.

13 Building and Distribution

13.1 Build Requirements

- Python 3.9+ with pip
- Logo PNG file at the path specified in build_exe.bat
- ICO file for taskbar icon (convert PNG to ICO at convertio.co/png-ico)

13.2 Build Command Summary

The build_exe.bat script runs the following PyInstaller command:

```
pyinstaller --onefile --windowed --name "BiblioBlitz_v3.2"
  --icon biblioblitz.ico
  --add-data "path\to\logo.png;."
  --add-data "path\to\biblioblitz.ico;."
  biblioblitz.py
```

13.3 Distribution

- Upload BiblioBlitz_v3.2.exe to GitHub Releases (supports files up to 2 GB).
- Upload source files (biblioblitz.py, build_exe.bat, README.md) to the GitHub repository.
- Create a new Zenodo release to update the DOI record.

14 Citation

If BiblioBlitz is used in academic research, please cite:

> *Poddar, A. (2026). BiblioBlitz v3.2 — Global Open-Access Academic Paper Downloader. Zenodo. https://doi.org/10.5281/zenodo.20469653*

BibTeX entry:

```
@software{biblioblitz2026,
  author    = {Poddar, Ayanava},
  title     = {BiblioBlitz --- Global Open-Access Academic Paper
Downloader},
  year      = {2026},
  publisher = {Zenodo},
  version   = {3.2},
  doi       = {10.5281/zenodo.20469653},
  url       = {https://doi.org/10.5281/zenodo.20469653}

}
```

15 Acknowledgements

BiblioBlitz uses the following free and open academic APIs and open-source libraries. The author gratefully acknowledges their developers and maintainers.

Resource	**Provider**	**URL**
CrossRef API	Crossref	https://api.crossref.org
OpenAlex API	Our Research	https://openalex.org
Semantic Scholar API	Allen Institute for AI	https://api.semanticscholar.org
PubMed / Entrez API	NCBI / NIH	https://eutils.ncbi.nlm.nih.gov
CORE API	Open University (UK)	https://core.ac.uk
Unpaywall API	Our Research	https://unpaywall.org

Resource	**Provider**	**URL**
CustomTkinter	Tom Schimansky	https://github.com/TomSchimansky/CustomTkinter
Pillow	Python Imaging Library fork	https://python-pillow.org
PyInstaller	PyInstaller development team	https://pyinstaller.org